Conscious artificial intelligence and biological naturalism

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Abstract

As artificial intelligence (AI) continues to develop, it is natural to ask whether AI systems can be not only intelligent, but also conscious. I consider why some people think AI might develop consciousness, identifying some biases that lead us astray. I ask what it would take for conscious AI to be a realistic prospect, pushing back against some common assumptions such as the notion that computation provides a sufficient basis for consciousness. I'll instead make the case for taking seriously the possibility that consciousness might depend on our nature as living organisms – a form of biological naturalism. I will end by exploring some wider issues including testing for consciousness in AI, and ethical considerations arising from AI that either actually is, or convincingly seems to be, conscious.

1.0 Introduction

As artificial intelligence (AI) systems gain competence and capability, a question arises about whether such systems might become (or might already be) conscious, as well as intelligent. A definitive answer to this question is not currently possible, given the lack of a consensus view about the minimally sufficient conditions for consciousness. The idea that AI could be or become conscious is nonetheless a widespread opinion (Butlin et al., 2023; Chalmers, 2023; Dehaene et al., 2017). How we think about the prospects for conscious AI matters. It matters from the perspective of the AI systems themselves, and it matters from the perspective of how human beings and human society are affected by systems that are, or appear to be, conscious.

In this paper I will address the prospects for creating conscious AI, and the pitfalls that accompany AI that is, or appears to be, conscious. I will start by asking why people might think that AI is on a trajectory towards consciousness, identifying some psychological biases that can lead to overestimation of the likelihood of this happening. I will then critically examine some of the assumptions that underpin the possibility of conscious AI, focusing on computational functionalism and substrate neutrality. I intend this discussion to give reasons to doubt that AI is, or could be, conscious. It is not intended to show that it is impossible to build a conscious artefact – but that such artefacts, if they are possible, may be closer to biological systems than assumed by many proponents of conscious AI. I will then move on to *biological naturalism*: the idea that consciousness is a property of only, but not all, living systems. I outline and motivate several variations of this view, recognising that 'life' need not be carbon-based, and without relying on any neo-vitalistic magic. If these arguments are on

track, conscious AI will need to be 'living AI'. The paper will finish by discussing wider issues, including how to test for consciousness in AI, the potential importance of embodiment and embeddedness, and concerns surrounding AI systems that are, or appear to be, conscious. Even if the prospects for real artificial consciousness remain remote, there are significant ethical and societal concerns relating to AI systems that give the powerful and perhaps cognitively impenetrable appearance of being conscious.[1]

1.1 Intelligence and consciousness

To begin, we need working definitions of both consciousness and intelligence. Consensus definitions, like consensus theories, are lacking for both these terms. But some ground rules are necessary.

I take intelligence to mean, broadly (and without moral connotation), *doing the right thing at the right time*. A more specific definition, drawn from a survey of many different definitions, is the "ability to achieve goals in a wide range of environments" (Legg & Hutter, 2007) (p.23). Intelligence in these senses, as in most or all other senses, has to do with the functional capabilities of a system. Reasoning, goal-directed planning, and linguistic competence are all aspects of intelligence, but non-human animals that lack these competences may still exhibit intelligent behaviour. More general aspects of intelligence may include the ability to flexibly respond to challenges, and to learn from experience. An ant that flexibly learns to find its way home through a desert landscape is arguably displaying intelligent behaviour (Simon, 1988).

AI systems are artificial systems that exhibit functional properties associated with intelligence. Unless otherwise qualified, I assume a standard view in which AI systems (or 'agents') are defined in terms of information processing architectures implemented in a computer, and potentially involving real or virtual interactions with (e.g., robotic) bodies and environments (Russell & Norwig, 2003). 'Artificial General Intelligence' (AGI) is the term typically given to (still hypothetical) AI systems that have attained or exceeded human-level intelligence in (at least) all the ways humans are supposedly intelligent. These systems are general in the sense of generalising beyond specific task domains and training regimes (Goertzel & Pennachin, 2007).

Consciousness is not the same thing as intelligence. Informally, consciousness is what goes away under deep general anaesthesia, or in dreamless sleep. More formally, a useful general definition comes from Thomas Nagel: "for a conscious organism, there is something it is like to be that organism" (Nagel, 1974). That is, it 'feels like' something to be a conscious system – there is a conscious experience happening – whereas it doesn't feel like anything to be an unconscious system – there is no conscious experience happening. Here, 'feeling' doesn't necessarily involve emotional content: any kind of conscious experience will do. It (probably) feels like something to be a bat, and it (probably) doesn't feel like anything to be a stone. I take it that there is a fact-of-the-matter about whether something is conscious, and that being conscious is not determined by social or linguistic consensus [see (Shanahan, 2024) for discussion].

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[1] See my section in (LeDoux et al., 2023) for a very short summary of the main points.

I treat consciousness as synonymous with awareness. I avoid the term 'sentience', which is sometimes used synonymously with consciousness/awareness, but which is more prone to confusion. Some people use sentience to denote mere sensitivity or responsiveness to an environment, without implying the presence of conscious experience (Kagan et al., 2022; Kagan et al., 2023). Others use sentience to mean a form consciousness that is necessarily valenced, where experiences feel 'good' or 'bad' in some sense (Thompson, 2022). The ambiguity over different interpretations of sentience adds unnecessary confusion.

Given these broad definitions, intelligence and consciousness are evidently distinct. In principle, a creature or system could do the right thing at the right time without there being something it is like to be that creature or system, and vice versa. However, in practice, these properties do not seem to be independent. It might be that consciousness is useful or even necessary for some forms of intelligence, and that some forms of consciousness require corresponding forms of intelligence.[2]

2.0 Why might we think that AI could be or become conscious?

The idea of conscious AI has deep roots in human culture. In 16th Century Jewish folklore, a golem made of clay – called Josef, or Yoselle – became conscious, and disturbingly violent, when activated by a magical incantation. More recently, science fiction has been a reliable source of imagined conscious creations, from HAL in Stanley Kubrik's *2001* (released in 1968), to Ava in Alex Garland's *Ex Machina* (released in 2014) and beyond.

Uniting these examples is that the artificial creations are intended by their creators to be humanlike, not necessarily in appearance (e.g., HAL), but in mind. This exposes a mixture of three embedded and intertwined psychological biases: anthropocentrism, human exceptionalism, and anthropomorphism. The case I make here is that these biases lead to intelligence and consciousness being conflated, which in turn drives an unwarranted optimism about the prospects for conscious AI.

Anthropocentrism is the tendency to place humans at the centre of things and to interpret the world in terms of human values and experiences. This is closely related to human exceptionalism – the tendency to see human beings as distinct from and superior to other forms of life. Putting these together, there is the temptation to take aspects of intelligence that seem distinctively and perhaps exclusively human – such as language and other 'advanced' cognitive abilities – as bound up with other supposedly distinctively humanlike qualities – such as consciousness. We humans know we're conscious, we think we're intelligent, and we feel we're special, so we assume that intelligence and consciousness go together. This way of thinking traces back at least to Descartes, for whom the immaterial,

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[2] For example, the ability to experience regret, as compared to mere disappointment, plausibly requires the ability to reason about the future consequences of actions and to decide between competing action plans. In general, whether intelligence and consciousness look independent will depend on whether one asks the question logically (where anything logically coherent is permitted), nomologically (given the laws of nature as they are), or in terms of systems – biological or otherwise – that actually exist. It seems plausible that the properties are logically and nomologically independent.

rational mind both endowed humans with the kind of consciousness that mattered, and separated humans from all other animals (Shugg, 1968).

Anthropomorphism is the complement to anthropocentrism: it is the tendency to project humanlike qualities into things that don't (necessarily) have them. For example, we are more likely to attribute emotions to things with faces than to things without faces (Alais et al., 2021). This can be useful when it helps us explain and predict a system's behaviour, as in Dennett's 'intentional stance' (Dennett, 1987), but it can be problematic when the unwarranted attribution of humanlike properties leads to poor explanations and/or unreliable predictions. When coupled with anthropocentrism and human exceptionalism, anthropomorphism can engender a tendency to attribute consciousness to things that seem to display those humanlike qualities that we think of as most distinctive, such as language and 'higher' cognition.

These biases go some way towards accounting for the view that AI is on a royal road to consciousness: that as machines get smarter, they will eventually cross a threshold at which they will also become conscious. Once a system gains a property we (anthropocentrically) feel is distinctively human, we are more likely (anthropomorphically) to attribute it with other supposedly distinctive properties, such as consciousness. The contribution of human exceptionalism is highlighted when the relevant threshold is assumed, implicitly or explicitly, to be the point at which General AI – the intelligence level characteristic of a human – is reached. And the pedestal on which we put language, as a canonical expression of distinctively humanlike intelligence, explains why the current wave of large language models (LLMs) has been so powerful in seducing our intuitions (Colombatto & Fleming, 2023).

There is arguably a fourth ingredient: the technophilic temptation to see ourselves at the cusp of a major transition: a Kurzwelian Singularity beyond which AI bootstraps itself beyond our understanding and control and into full-fledged machine consciousness (Kurzweil, 2005). This temptation is motivated in part by the psychological difficulty of grasping the nature of exponential change (Wagenaar & Sagaria, 1975). From any point on an exponential curve the future looks impossibly steep, while the past looks irrelevantly flat. Another part comes from a posthuman desire to transcend the limits of biological mortality by uploading one's consciousness into the pristine circuits of some future supercomputer and therefore to live – or at least to 'be' – forever. Indeed, the idea that we already exist within a computer simulation has received philosophical attention (Bostrom, 2003).

"post-human" is sometimes used to refer to a time when human intelligence won't be that important – what will be the value of human consciousness, emotion, and other things that make us human? Can these be quantified like intelligence?

Altogether, it is hardly surprising to witness the widespread emergence of the view that AI will inevitably, and perhaps very soon (see ingredient number four, above), lead to real artificial consciousness.[3] But having recognised the biases driving this view, we are freed up

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[3] This view is particularly apparent in (even relatively well informed) media coverage. This is from a *New Statesman* piece on Open AI, in late 2023: "Company documents state that the board's purpose is to decide when the company has developed 'highly autonomous' artificial general intelligence (AGI) that 'outperforms humans at most economically valuable work' and to ensure that this technology 'reshapes society' in a manner that is safe and benevolent. *In other words, the board exists to decide when the machine is sentient – when it is alive – and to ensure that the new consciousness in moral*". My italics.
https://www.newstatesman.com/business/2023/11/openai-sam-altman-chatgpt-executed-priests

from assuming that making machines smarter will inevitably make them conscious, even though we might well become more susceptible to judging them as being conscious.

if we treat them as conscious, does it matter if they are?

None of this means that conscious AI cannot happen. One could try to develop conscious AI deliberately, rather than assuming it will come along 'for free' with increasing intelligence. I will return to this strategy later. But for conscious AI to happen – whether through the unlikely trajectory of simply increasing intelligence, or through some other way – it must be the case that consciousness is the kind of thing that AI can in principle have. This brings us to the assumption of *computational functionalism* – an assumption which underpins the very idea of conscious AI.

3.0 Computational functionalism

Computational functionalism is the view that computations of some kind are sufficient to instantiate consciousness. It derives from the popular philosophical notion of functionalism, which says – roughly – that consciousness is a matter only of what a system does, not of what it is made out of (Putnam, 1975).

what about why it does things?

More specifically (though still very briefly) the arguments run as follows. Functionalism in the philosophy of mind proposes that a system has a mind when it has a suitable functional organization (Putnam, 1967). Mental states play roles within this functional organization, and are characterised by their interactions with sensory inputs, motor outputs, and other mental states. Applied to consciousness, functionalism proposes that it is a property of the functional organisation of its underlying mechanism, which for human beings is the (embedded and embodied) brain. *Computational* functionalism is the further claim that the kind of functional organisation that matters for mind in general, and for consciousness in particular, is computational in nature: that it consists in computational parameters, together with inputs and outputs (Putnam, 1988).

Questions about whether (computational) functionalism holds are distinct from questions about whether consciousness has a function or functions, in the sense of playing a useful role for or within a system. The ideas are nonetheless related: functionalism provides an appealing framework within which to address questions about the functions of consciousness.[4]

Functionalism, and computational functionalism, are widely accepted both explicitly and implicitly. In a widely-read review of prospects for conscious AI, Patrick Butlin, Robert Long, and colleagues confine themselves to theories and models that assume this position, and explicitly acknowledge that their conclusions depend on whether computational functionalism holds (Butlin et al., 2023). Implicitly, any talk of the mind or consciousness in terms of 'information processing' assumes some kind of computational functionalism, since information processing is what computers do.

what else would the brain do if not compute?

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[4] See (Cleeremans & Tallon-Baudry, 2022; Seth, 2009) for more on the functions of consciousness, and (Piccinini, 2023; Rescorla, 2020) for thorough treatments of functionalism and computational theories of mind more generally.

Functionalism and computational functionalism are also often conflated, as evidenced by the prevalence of information processing language throughout cognitive science and the neurosciences. However, although computational functionalism entails functionalism, the reverse is not true. Forms of functionalism are possible which are non-computational in nature (Piccinini, 2018, 2023).

Just as psychological and historical factors drive associations between intelligence and consciousness, there is a wider context which helps explain the appeal of computational functionalism. Recognising this context helps assess its plausibility.

Functionalism in general was motivated in part by the perceived inadequacies of the dominant metaphysical views of dualism (broadly, separate realms for the mental and the physical) and materialism (broadly, consciousness is property of matter, suitably arranged) (Shagir, 2010). The focus on functional organisation rather than immaterial essences, or supposedly mysterious properties of the material substrate, provided an attractive middle ground. It also provided an appealing framework for thinking about consciousness in brains of differing physical structure, both within and across species. But being more appealing than dualism or naïve forms of materialism doesn't mean being right.

Functionalism, and computational functionalism in particular, was also motivated by the rapid progress of computer science and early forms of AI in the second half of the 20th Century (Boden, 2008; Dupuy, 2009). These gains helped embed the metaphor of the brain as a computer, normalise the language of information processing as describing what brains do, and arguably galvanise the discipline of cognitive science. But metaphors are in the end just metaphors (Cobb, 2020). The fact that mental processes can sometimes be usefully described in terms of computation is not a sufficient basis to conclude that the brain actually computes, or that consciousness is a form of computation.

For conscious AI to be possible for the contemporary and conventional forms of AI noted above, or for extrapolations of this technology, it needs to be the case that the relevant computations provide a sufficient basis for consciousness. This requires both that computational functionalism holds, and that AI can implement the relevant computations. And the fact that computational functionalism is so frequently assumed does not mean it is true. After all, as science writer Oshan Jarrow neatly puts it, "if we keep making the assumption without returning to it, the question itself begins to disappear"[5].

3.1 *Substrate neutrality and neural replacement*

One way to put the relationship between conscious AI and computational functionalism is in terms of *substrate neutrality*. Substrate neutrality or (equivalently) substrate independence is the claim that the same kind of mental states (including conscious states) can occur in systems with different physical substrates – i.e., systems made out of different kinds of stuff. Computational functionalism is intimately related to substrate neutrality. If computational functionalism holds, then at least some substrate neutrality follows, for those substrates able to implement the relevant computations. And if substrate neutrality holds, then computational functionalism gains support because it provides an account of mental states that does not depend on the material properties of a substrate, beyond their ability to implement the relevant computations. Conscious AI requires substrate neutrality to hold at least for the alternative substrate in question: silicon.

One of the most common supports for substrate neutrality in this context is the 'neural replacement' thought experiment (Chalmers, 1995; Haugeland, 1980). The basic scenario is that a person's brain cells[6] are replaced, one by one, with silicon alternatives. Each silicon brain cell exactly replicates the input/output function of its biological counterpart, in all situations. Eventually, the person's brain has only silicon parts, yet its functional organisation is entirely preserved, and so – from the outside – the person would behave exactly as before. If replacing just one brain cell doesn't make a difference to the person's consciousness – and this seems unlikely – then why should replacing 100, or indeed all of them? Would their consciousness simply fade away? And, if so, is it plausible that their behaviour could remain unchanged while conscious experience completely disappears? The way out of these strange implications seems to be to accept substrate neutrality.

This thought experiment can be criticised from several directions. (Block, 2019) points out that it begs the question. If consciousness is substrate dependent, then something would indeed happen to the person's conscious experience as the neural replacement progressed. [This doesn't mean the person need notice this happening: change blindness experiments suggest that change of experience does not always entail the experience of change (Simons & Levin, 1997)].

↳ they may think they're still conscious but that doesn't mean they still are

Other criticisms (also mentioned by Block) point to plausibility considerations. Building on early arguments from (Bechtel & Mundale, 1999), Godfrey-Smith argues that as living components (neurons) are replaced by non-living (e.g., silicon) parts, there will inevitably be differences in how the system works on the inside, and in its overall behaviour (Godfrey-Smith, 2016). These differences emerge because of influences on neural activity arising from metabolic constraints, electromagnetic fields, fine-grained timing relations, and potentially other factors. If the system behaves differently, internally and externally, then there is no pressure from this argument to suppose that consciousness need be maintained.

Rosa Cao extends this line of argument, providing many descriptions of how mental functions are thickly intertwined with each other and with their underlying physiological and metabolic bases (Cao, 2022). She argues that because mental functions arose through evolution, the realizers of these functions have accumulated many dependencies and contingencies, some of which may seem arbitrary, and which collectively lead to what (Wimsatt, 1986) called a 'generative entrenchment' in the internal organisation of the brain. This generative entrenchment imposes many constraints on possible alternative implementations, which likely exclude alternative substrates such as silicon.

→ if consciousness were to be achieved through silicon, it would undergo its own evolution (we could simulate evolution too but this argument is concerned with entrenchment that arrives naturally).

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[5] https://www.vox.com/future-perfect/351893/consciousness-ai-machines-neuroscience-mind

[6] I say 'brain cell' rather than 'neuron' because, even if computational functionalism holds, it might be that things other than neurons are needed to implement the required computations. Glial cells, neurotransmitters, and other biochemical gubbins may in principle all play important computational or functional roles. Chalmers also includes axons and other aspects of neural connectivity in his version of this thought experiment.

Many examples of these kind of constraints can be given. One from Cao (2022) is about the role of nitric oxide, a neurotransmitter involved in many neuronal functions. Nitric oxide is distinctive because it freely diffuses across cell membranes, meaning that any silicon replacement would need to be able to detect this substance (or a simulated version of it) at an infeasible number of (perhaps simulated) locations. Another comes from a recent study showing that some neuronal spikes serve a homeostatic role by protecting neurons from the build-up of damaging reactive oxygen species (Chintaluri & Vogels, 2023). This means that the spiking activity of a neuron, generally considered key to its functional role, cannot be decoupled from its metabolic foundations. Even chemotactic behaviour in bacteria may be inseparable from metabolic processes (Alexandre, 2010; Egbert et al., 2010), suggesting a link between metabolism and mind that I will return to later.

→ could homeostasis (some mental or emotional form) also be a feature of consciousness? what could define silicon homeostasis?

Derek Shiller raises concerns at a more abstract level (Shiller, 2024). He starts from the worry that a liberal interpretation of functionalism could attribute mental states and consciousness to all kinds of things where these properties are clearly absent, by carving things up in particular ways. For example, a bucket of water may transiently reflect the functional organisation of a brain, if the water is 'carved up' into parts in the right way.[7] To prevent this, it is necessary to place constraints on what can count as a system's parts. Shiller identifies several constraints, such as the need for parts to play their functional roles in virtue of intrinsic causal powers. He argues that these constraints are not satisfied by current neural network models, and that they may be difficult to satisfy for non-biological substrates in general.[8]

These plausibility arguments do not disprove substrate neutrality, but they do challenge assumptions – based on thought experiments like neural replacement – that it must be the case. Brains seem to be the kind of thing for which it is hard and perhaps impossible to separate *what they do* from *what they are*.[9]

These arguments also apply beyond substrate neutrality to the more general notion of *multiple realisability*. This is the claim that the same kind of mental states (including conscious states) can occur in systems with different physical implementations, but not necessarily different substrates (Putnam, 1967). Multiple realisability evidently holds to some degree, to the extent that non-identical brains can implement similar (conscious) mental states. But it need not hold in general, for arbitrary physical implementations. It is, however, the challenges to substrate neutrality that most directly undermine the plausibility of conscious AI.

→ non-identical LLMs often produce similar results, and identical LLMs produce different results since they're non-deterministic. this is certainly not unique to human consciousness but worth considering a bit more.

→ unrelated: ironically, perhaps the ability of the mind to wander?

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[7] This example is from Ian Hinckfuss (Shiller, 2024).

[8] These various objections to substrate neutrality are meant in a nomological sense rather than in a logical sense. They also apply to Chalmers' 'dancing qualia' argument, where instead of neurons being replaced, control of a human agent is switched back and forth between the real biological brain and a (presumed) functional duplicate installed in a system with a different physical constitution. See (Chalmers, 1995; Godfrey-Smith, 2016).

[9] One response to these plausibility arguments is to dismiss them as merely practical. To say something like: "Of course it's difficult (to build a silicon brain replicating everything that matters for mental states) but let's just imagine we can." But one cannot draw useful conclusions from 'just imagining' situations for which there are good reasons to believe cannot (nomologically) be the case. This is a weakness with conceivability arguments in general. The (in)famous zombie argument suffers similarly (Chalmers, 1996; Seth, 2021).

3.2 Mortal and neuromorphic computation

A related perspective involves the concept of 'mortal computation' (Hinton, 2022; Ororbia & Friston, 2023). A core principle of standard computer science is that software should be separable from hardware, so that the same algorithm executed on different hardware would give the same result. One should be able to take the weights from a neural network trained on one computer and implement the trained network on another, to get the same behaviour. This is what happens in practice, which is why software is useful.

In this sense, software is 'immortal' because its existence and utility doesn't depend on the continued existence of any specific instance of hardware. But immortal computation is expensive. It requires continual error correction to ensure that 1s typically remain 1s (and 0s remain 0s). As algorithms and models grow in complexity, which is what we are seeing in parameter counts for contemporary AI systems based on deep neural networks, the computational and energetic costs of error correction, and therefore of computational immortality, grows quickly.

An implication of this argument is that biological brains, which are highly energy efficient, cannot be implementing immortal computations. If they are implementing computations at all (which is not obvious, as I will discuss next), then these computations are likely to be *mortal*, which means they cannot be separated from the 'hardware' (or 'wetware') which implements them. This in turn places constraints on the multiple realisability and substrate neutrality of these (mortal) computations. In particular, the degree of substrate neutrality required for conscious AI is unlikely to hold because (conventional) AI is based on an implementation paradigm assuming computational immortality. I find this a particularly powerful argument against the plausibility of conscious AI because it is based on limits that arise from adopting a computational view of mind.[10]

→ to look up: neuroplasticity and how the brain's physical structure changes with time, challenge, environment, etc

Of course, challenges to the standard computer science paradigm have been around for a long time. Proponents of *neuromorphic computing* have long argued for alternative forms of computation based on hardware that has more in common with biological neural networks, compared to the CPUs and GPUs that dominate modern computers (Mead, 2020; Schuman et al., 2022). Neuromorphic chips have been developed which use spikes to communicate between different hardware elements, potentially offering substantially greater energy efficiency, among other benefits (Merolla et al., 2014). Some neuromorphic approaches emphasise analogue rather than digital computation, tying function more directly to substrate (Douglas et al., 1995), and in the limit challenging the validity of 'computation' as a description of the dynamics in play. It is plausible that the more neuromorphic a chip becomes, the less likely substrate neutrality is to hold.[11] I prefer to think of neuromorphic

continuous vs. discrete 0s/1s

→ if more neuromorphic = better analogue for a biological brain, then doesn't it become more substrate neutral? I understand that the human brain does not necessarily uphold multiple realisability, and neither do these chips, but why does that diminish substrate neutrality?

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[10] (Kleiner, 2024) makes a similar point, arguing that if computational functionalism is true, then consciousness requires mortal computation, and so AI cannot be conscious. Kleiner's argument is similar to (Shiller, 2024) in focusing on how an unconstrained (computational) functionalism can lead to an overly liberal attribution of consciousness. While Kleiner and I reach similar conclusions we do so via different routes. Kleiner also proposes a proof that AI can't be conscious based on assuming computational functionalism, whereas I do not wish to assume computational functionalism. See also (Kleiner & Ludwig, 2023) for another argument against conscious AI that doesn't assume computational functionalism.

[11] A fascinating example comes from early work by Adrian Thompson at Sussex University, using genetic algorithms to optimise the configurations of field programmable gate array (FPGA) chips to perform various

computing as offering an alternative to conventional AI, rather than as mounting a conceptual challenge to computational functionalism in the way that mortal computation does.

3.3 *Emergence and the separation of scales*

The notion of substrate neutrality can also be usefully connected to hierarchical emergence in physics. Physicists speak of a 'separation of scales' which allows laws to be formulated at higher – emergent – levels of description, that do not require knowing the details at finer-grained levels of description. Engineers can build bridges without knowing about quantum mechanics. Computer programmers can write software without worrying about implementation details. The substrate neutrality (and immortality) of software rests on this strong separation of scales.

Recently, empirical measures have been developed which formalise notions of 'informational closure' and 'causal closure' which underwrite the separation of scales in complex systems [e.g., (Barnett & Seth, 2021; Rosas et al., 2024)]. A macroscopic (higher-level) variable – or 'coarse graining' – is casually or informationally closed when interventions on it (causal closure), or observations of it (informational closure), are sufficient to determine or predict the higher-level outcome(s) to the extent that these outcomes are determinable or predictable. To put it another way: knowledge of (or interventions at) lower, microscopic levels do not add anything to macro-level predictability, or help determine macro-level outcomes, beyond what can be done at the macro-level. This is guaranteed by-design for (properly functioning) computers. But it would be a mistake to assume that these forms of closure hold for complex systems generically. In particular, thanks to the generative entrenchment inherent in evolutionary design, they may be partially or wholly absent in biological systems.

→ to recap: causal and informational closure are the mechanisms that allow for emergent hierarchies

Although there is no direct challenge to substrate neutrality here, the ability to measure causal and informational closure helps make clear that substrate neutrality occupies only a fraction of available design space. These measures may also help empirically determine whether, or to what extent, neural dynamics can be abstracted away from finer-grained levels of description.

3.4 *Beyond computational functionalism*

The arguments in this section do not disprove computational functionalism. But they do render it less plausible, and less appealing, by suggesting limitations on substrate neutrality and multiple realisability. Certainly, they give good reasons for not assuming that

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functions. In some of his studies, he found that the optimal configurations implicitly made use of spatial relationships between the FPGA components and the chip's power supply. The function of the chip therefore depended on its precise spatial layout, severely constraining multiple realisability. Other FPGA configurations turned out to depend on ambient room temperature – another constraint on multiple realisability that Adrian creatively tried to avoid by optimising his FPGAs in a range of fridges set at different temperatures (Thompson, 1998). I remember these fridges appearing in our lab one day, while I was studying for my doctorate.

computational functionalism and substrate neutrality must be the case.[12] Since conscious AI depends on computational functionalism holding, and on substrate neutrality extending at least to silicon – let's call this *sufficient substrate flexibility* – there are also good reasons to question whether conscious AI is possible, again for AI understood as computations implemented on a silicon substrate.

Considering what kind of thing a brain is, it is perhaps surprising that computational functionalism became so pervasively adopted. When we look inside a brain, we do not find anything like a sharp distinction between 'mindware' and 'wetware' of the kind we find between hardware and software in a computer. Brain activity patterns evolve over multiple scales of space and time, continuously influenced by, and influencing, the chemical diffusion of neurotransmitters, the physical structure of the neural networks themselves, and various properties of cellular infrastructure, all deeply interwoven with a molecular storm of metabolic activity. A single neuron is a complex biological system in and of itself, busy maintaining its own integrity and regenerating the conditions for its own material existence. The metaphor of the brain as a computer is an increasingly inconvenient fiction, and it seems myopic to consider all this multiscale biological activity as simply an enabling condition for the right kind of computation.[13]

This line of thought gives rise to other possibilities. One possibility is that consciousness may still depend on patterns of functional organisation, but not on computations. This position is *non-computational functionalism* (Piccinini, 2018, 2020; Prinz, 2012). It is well aligned with the large body of work that attempts to understand brain activity using formalisms that do not assume a computational stance, such as the influential 'dynamical systems' approach to cognitive science (Van Gelder, 1995). This approach provides various mathematical tools and methods – usually based on differential equations – to model how properties of embodied and embedded neural systems change over time.[14] Dynamical systems theory approaches historically did not focus on consciousness and in this regard are complementary to non-computational functionalism. Something similar can be said of the various perspectives under the banner of '4E' cognitive science, which emphasises embodied, embedded, enactive and extended aspects of brain-body-world interactions – though consciousness appears more often in work of this kind (Cosmelli et al., 2007; Stewart et al., 2010; Thompson, 2022). These views differ in important details, but they share a disdain for the idea that the brain is a computer, and that the mind is essentially computational.

computational
systems = discrete,
binary, etc

dynamical =
continuous, real-time
etc

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[12] Computational functionalism has come under fire from other directions too, for example through philosophy of mind considerations about intentionality (Shagir, 2010) and the distinction between syntax and semantics (Cole, 2023). I do not rely on these arguments here. But they played at least some part in one of the architects of this view, Hilary Putnam, repudiating his own brainchild (Shagir, 2010). Another line of attack on computational functionalism, also not relied on here, comes from Roger Penrose's controversial argument from Gödel's theorem (Penrose, 1989).

[13] One way to rescue a form of computational functionalism might be to appeal to the idea that everything in the universe is a form of computation (Wolfram, 2002). This seems a high price to pay, and when combined with computational functionalism might entail a form of panpsychism. Note that the claim that *computation* is fundamental is distinct from the claim that *information* has ontological status (Wheeler, 1989). One can accept the latter without committing to the former.

[14] See again (Kleiner & Ludwig, 2023) for an argument connecting dynamical aspects of neural activity to the implausibility of conscious AI.

Various neuroscience-based theories of consciousness emphasise or can be framed in terms of dynamical rather than computational elements. These include classic works by Walter Freeman and Rodolfo Llinas linking conscious perception to attractor dynamics in neural systems (Freeman, 1999; Llinas et al., 1998). Early theories that emphasised neural synchronisation, for example in the gamma frequency band (Crick & Koch, 1990), also fall into this camp, as does the 'dynamic core' hypothesis (Tononi & Edelman, 1998) which highlighted integration and differentiation as key neurodynamical properties underlying consciousness – and which later resurfaced as the integrated information theory of consciousness (Albantakis et al., 2023; Tononi et al., 2016).[15] Predictive processing and the free energy principle (Clark, 2019; Friston, 2010), which I will expand on later, may also qualify, depending on how these theories are interpreted (see Section 4.2). More generally, the tradition of neurophenomenology, inspired to a significant degree by the work of Francisco Varela, often appeals to non-computational dynamical constructs as bridging principles between neural processes and phenomenological properties (Thompson & Varela, 2001; Varela, 1996).[16]

Dynamical neuroscientific and (neuro)phenomenological approaches to consciousness are promising. However, it seems fair to say they are in general agnostic about whether realising the dynamical principles proposed by a given theory, in some alternative material, would be sufficient to instantiate consciousness. (I hesitate to use the term 'conscious AI' here because we are no longer talking about implementing computations, nor are these approaches generally focused on intelligence.) The extent to which dynamical theories distinguish between *realising* a principle or mechanism, and merely *simulating* it, is also not clear, and may vary from theory to theory. The distinction is important because one would have to assume computational functionalism to assume that simulation and realisation (of consciousness) would be equivalent. Realising non-computational dynamical processes in alternative substrates again raises the challenge of substrate neutrality (or sufficient substrate flexibility), and it does so in a more obvious and pressing way since the in-principle substrate neutrality provided by computationalism cannot be appealed to.

3.5 Interim summary

Conscious AI requires both computational functionalism to hold, and sufficient substrate flexibility such that the computations sufficient for consciousness can be implemented in AI hardware (silicon). But there are good reasons for thinking that the functions or computations implemented by (conscious) biological systems may not be separable from their material basis. This means that the substrate flexibility required for conscious AI may not hold. There are also alternatives to assuming that brains compute, or that consciousness is a matter of computation, undermining unmotivated assumptions of computational functionalism.

The arguments so far remain compatible with a functionalist perspective on consciousness, albeit a pared back functionalism shorn of its usual computational connotations. But there is a second possibility, which is that the biological basis for consciousness matters in a more fundamental way than just as serving as an enabling factor for the right kind of dynamics or computation. The intuition here is that the biological substrate matters for consciousness in a similar way to how rain depends on the presence of water molecules: no matter how detailed a simulation of a weather system is, it never gets wet inside the simulation.[17]

This brings us to *biological naturalism*: the perspective that consciousness is a biological phenomenon. In the next section I will make a positive case for various forms of biological naturalism, framed in terms of explanatory links between the properties of mind and consciousness, and properties of their biological underpinnings. This case rests on a particular theoretical perspective on perception, cognition, and consciousness. I will make it from two directions, meeting in the middle: the story from predictive processing, and the story from the free energy principle.

4.0 Towards biological naturalism

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[15] For other dynamical perspective on consciousness see (Cosmelli et al., 2007; Edelman, 1989).

[16] The fact that these theoretical perspectives often make use of computational models does not mean they assume any form of computationalism as a theory of mind or of consciousness. Computational models provide powerful general-purpose methods for simulating all kinds of systems, computational or not. A standard example, from outside neuroscience, is provided by computer models of weather systems. This point is particularly relevant for the emerging discipline of 'computational (neuro)phenomenology' which uses computational methods to give precision and explanatory power to phenomenological and dynamical concepts which may be non-computational. See, for example, (Ramstead et al., 2022).

Biological naturalism is part of a family of ideas that stress intimate connections between properties of life and properties of mind, including consciousness. It says that consciousness is a biological phenomenon that arises from neural and perhaps other biological processes in the brain and body (Searle, 2017). Importantly, biological naturalism is distinct from biopsychism. The former says that consciousness is a property of only, but not all, living systems. The latter, that it is a property of only *and all* living systems (Haeckel, 1892; Thompson, 2022). I am concerned here only with biological naturalism, and – specifically – with making the case that some properties of life may be necessary for consciousness.

→ this means we can look to existing examples of systems that are living but not conscious

John Searle, who coined the term, based his defence of biological naturalism partly on a rejection of computational functionalism motivated by – among other things – his famous 'Chinese Room argument' (Searle, 1980). I take this argument to be primarily about the distinction between syntax and semantics and the possibility for true (strong) AI, and only secondarily about consciousness (recall that intelligence and consciousness are distinct). I will therefore not dwell on it, and turn instead to the positive case.[18]

→ syntax = structure
semantics = meaning

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[17] A common retort goes like: "But wouldn't a simulated rainstorm feel wet for a simulated agent within the simulation?" The idea in this objection is that whether or not we recognise a simulated rainstorm as simulated depends on whether we're experiencing the storm from inside or outside the simulation (Chalmers, 2022; Hofstadter, 1981). The objection doesn't work because it already presupposes that computation/simulation is sufficient for experience; see also (Wiese, 2024).

→ but who decides? nobody is able to experience being both inside and outside the simulation, how can we say it isn't wet!

[18] A brief diversion may nonetheless be of interest. The Chinese Room argument runs as follows. A person who does not understand Chinese is locked in a room. She receives Chinese symbols from outside the room, which she translates into other symbols by following a rule book which is written in English, which she does understand. The argument goes that the whole system (room plus person plus rule book) can effectively answer questions in Chinese, despite no part of it understanding Chinese. Therefore, 'understanding' must involve more than mere symbol manipulation. There is a vast literature discussing this argument (Cole, 2023). The Chinese Room argument is primarily to do with intelligence and understanding. The connection to

4.1 *The story from predictive processing*

Predictive processing, as I use the term, labels a range of theories about perception, cognition and action (Clark, 2013; Hohwy, 2013; Hohwy & Seth, 2020; Rao & Ballard, 1999). These have in common the idea that perception is a form of probabilistic inference, and that the brain implements this inference through a process of prediction error minimisation.

→ sounds like LLMs so far

The standard predictive processing story is that perceptual content is not 'read out' from incoming sensory signals, but is instead constituted by the brain's 'best guess' about the causes of these sensory signals. In Bayesian terms, perceptual contents are constituted by approximately optimal posterior beliefs, given by a weighted combination of prior beliefs and sensory signals (likelihoods).[19] Since exact Bayesian inference is generally analytically intractable, the brain implements an approximation: prediction error minimisation. The idea is that the brain implements a hierarchy of generative models (in Bayesian terms, the joint distribution of the likelihood and prior) which are able to generate predictions about expected sensory signals, and which cascade down through perceptual hierarchies. These 'top-down' (or inside-out) predictions are compared with bottom-up (outside-in) sensory signals to form prediction errors, which are passed back up the hierarchy. By updating predictions to minimise prediction errors, this process settles on an approximately optimal posterior – a 'best guess'. Prediction errors can also be minimised by performing actions to bring about the expected sensory data – a process called active inference (Friston et al., 2017).[20]

→ iirc this is because Bayes thm is

$$P(\theta|x) = \frac{P(x|\theta)P(\theta)}{P(x)}$$

where $P(x)$ is intractable

The core claim from the perspective of consciousness is that conscious contents are given by the brain's 'best guesses' – by the joint content of the top-down perceptual predictions. This intuition is nicely captured by the notion of conscious perception as a 'controlled hallucination' – emphasising that conscious contents are actively generated, rather than passively registered (Seth, 2021).

→ to summarize: our brain predicts $P(\theta|x)$ where x (observation) is sensory signal and θ (parameter) is the perception

This standard story can be extended to account for conscious perceptions of the body, as well as of the world. These body-related perceptions arguably form an essential part of the experience of being a 'self'. Experiences of emotion and mood, which are central to selfhood, may depend on interoceptive inferences – probabilistic best-guesses about the sensory signals conveying information about the physiological condition of the body (Barrett & Simmons, 2015; Seth, 2013). Importantly, the functional role of interoceptive predictions is likely to be more concerned with physiological regulation than with discovering what's

I was thinking about how part of what it means to be conscious is to have a self ("I am"). Self implies feeling, which the

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consciousness seems to arise mainly via the intuition that understanding necessarily involves consciousness. There are many interesting issues hereabouts about whether understanding can be divorced from consciousness, perhaps through grounding AI systems in the physical world via embodied robotics. See (Pepperell, 2022) and https://theconversation.com/it-takes-a-body-to-understand-the-world-why-chatgpt-and-other-language-ais-dont-know-what-theyre-saying-201280.

[19] Note that belief in the Bayesian sense does not connote a (personal-level) psychological belief, as in 'I believe that Manchester United will win the Champions League'. Bayesian beliefs are probability distributions, usually characterised by a mean and a variance or precision.

[20] Many extensions to this simple predictive processing architecture are possible. In one recent extension – hybrid predictive coding – predictions and prediction errors flow in both directions, but at different time scales, corresponding to slow (iterative) and fast (amortized) inference (Tschantz et al., 2023). These different forms of perceptual best-guess may map onto different kinds of phenomenology.

author argues implies/requires a biological substrate.

there. This distinction can be framed in terms of instrumental versus epistemic inference. Epistemic inference is broadly about inferring the most likely causes of sensory signals, as in standard predictive processing, or – more generally – about discovering more about the world (and body). Instrumental inference is more about control than about discovery. In instrumental inference, prior beliefs can serve as targets or 'set points'. Minimising (interoceptive) prediction error through action can implement homeostatic or allostatic regulation relating to these set points (Seth & Friston, 2016).[21]

This control-oriented perspective on interoceptive inference inherits from long established cybernetic ideas emphasising the importance of prediction for control (Conant & Ashby, 1970; Seth, 2015). This suggests that the evolutionary driver for predictive processing as a general mechanism underlying conscious perception may have been a fundamental biological imperative for allostasis – for staying alive. It also sheds light on some relevant phenomenological distinctions. Visual experiences share, to a first approximation, a range of phenomenological characteristics including spatiality, object-relatedness, and so on. These characteristics are well aligned with (epistemic) predictions which have the broad function of figuring out 'what's there'. Emotional experiences have a different character: they are shaped primarily by valence – things feeling (variations of) good or bad. These characteristics are well aligned with control-oriented predictions, which have the broad function of allostatic regulation.

→ isn't the evolutionary driver for everything staying alive?

→ visual = what's out there, helps keep us safe
emotional = what's in here, keeps us stable

These observations together suggest a kind of epistemic continuity: a weak form of biological naturalism in which the physical basis of conscious experiences of world and self can only be understood in light of our nature as living, self-sustaining organisms. Perhaps there is even a minimal 'ground state' of conscious experience, which – instead of being entirely free of content – is characterised by an inchoate, shapeless, formless feeling of simply 'being alive'.[22] This putative ground state could then scaffold all other forms of conscious contents, whether self-related or world-related. Informally: we experience the world, and the self, *with*, *through*, and *because of* our living bodies (Seth, 2021).

→ I think there needs to be some classification of which living organisms experience consciousness, and what it is about their biology that allows this
→ addressed later on pg. 19 before 4.3: author will not address this

We need not stop here. A step towards a stronger form of biological naturalism can be taken by noticing that the imperative to 'stay alive' does not bottom out at any particular 'implementation level' in our biology. Even single cells – including neurons – are continuously engaged in maintaining their own physiological integrity. This homeostatic process can be directly linked to higher-level functional properties. One example was given earlier: neuronal firing patterns may partly depend on how apparently spontaneous spiking can prevent the build-up of damaging chemicals (Chintaluri & Vogels, 2023). If there is no clear boundary separating the substrate from the processes and dynamics that arise from it, then it is harder to motivate the intuition that the substrate does *not* matter.

More than this: living systems have 'skin in the game' (Aru et al., 2023). They are constantly engaged in processes of self-maintenance within and across multiple levels of organisation. The (higher-level) mechanisms of predictive processing are, on this view, inextricably

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[21] Allostasis is a generalisation of homeostasis that emphasises stability through change: for example, our blood pressure transiently rises as we stand up to prevent us from fainting (Sterling, 2012).

[22] For a different take on minimal phenomenal experience, see (Metzinger, 2024).

embedded in their (lower-level) biological context. This embedding is necessary for realising the higher-level properties, and for explaining their connection to consciousness.

We can go even further by appealing to the concept of *autopoiesis*, which derives from the Greek for 'self-production' or 'self-creation'. Autopoietic systems are special because they continually regenerate their own material components through a network of processes, and because this network of processes actively maintains a boundary between the system and its surroundings (Maturana & Varela, 1980). Biological cells are prototypical examples of autopoietic systems: they do not merely implement functions, but continuously regenerate their own material basis (self-production) and integrity (self-identification). Human-designed systems, like computers and factories, are generally not autopoietic. A factory takes in raw materials and produces some output – perhaps a car. The factory does not produce itself. A computer takes in some input and generates some output. It also does not produce itself. Systems like these are *allopoietic* – 'other producing'.[23] For autopoietic systems there really is no way separate what they are from what they do.

→ main argument: brain's self-sustaining and self-creating nature is what makes it distinct from a computer

→ i'm not sure this fully holds with LLMs but i can't delineate exactly why

The idea surfacing here is of a stronger, more ontological continuity between life and consciousness, in which the predictive processes that underpin all conscious experiences are inextricably tied to the autopoietic and excitable nature of living matter, and not merely to the patterns of functional organisation (computational or otherwise) that this matter enables.[24] The motivation for this idea rests on recognising that cellular autopoiesis can itself be understood as depending on – or consisting of – a kind of prediction error minimisation. This connection brings us into the territory of the *free energy principle*.

4.2 *The story from the free energy principle*

Having followed the predictive processing story first, the return journey will be faster. This journey starts from identifying basic constraints on the nature of systems that actively resist disorder so as to continue to exist – and from there recovers the entirety of the larger predictive processing view about consciousness.[25]

We begin with the autopoietic nature of living systems, whether cells or entire organisms. Unlike other kinds of things, like rocks, or computers, these systems actively (re-)generate their own material basis and maintain their boundaries over time. From the perspective of the free energy principle, this means that they actively resist the dispersion of their internal states – a dispersion otherwise mandated by the second law of thermodynamics. This in turn means they must exist in a state of low entropy, maintaining themselves out of

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[23] Are there examples of human-designed autopoietic systems? Synthetic biological systems would come closest. But these arguably fall short because the (autopoietic) component elements – cells – are at best modified by humans rather than created *de novo*.

[24] These varieties of continuity recall the 'mind-life' continuity thesis, which also has weak and strong versions. Weak versions propose that anything that has a mind is alive, while stronger versions propose that "mind is literally life-*like*" [(Godfrey-Smith, 1996), p.320, emphasis in original]. The distinction is therefore different from that between weak and strong forms of biological naturalism, not least because the mind-life continuity thesis does not say anything specifically about consciousness. See (Kirchhoff & Froese, 2017) for a discussion of mind-life continuity in the context of the free energy principle and autopoiesis.

[25] See chapter 10 in (Seth, 2021) for slower version of this journey, with more stops along the way. For more on the free energy principle itself, see (Bogacz, 2017; Buckley et al., 2017; Friston, 2010; Parr et al., 2022).

thermodynamic equilibrium with their environment, in order to fend off the inevitable descent into disorder. To stay alive means to continually minimise entropy because there are many more ways of being mush than there are of being alive. (It is sometimes said that staying alive involves resisting the second law. It is better to say that living systems take advantage of the second law, since it the transformation of low entropy fuel/food into high entropy products, through metabolism, that enables living systems to remain in non-equilibrium (quasi) steady-states[26]).

The challenge here is that entropy cannot be directly measured by the system itself, and so cannot be minimised. This is where free energy comes in. Free energy – technically the variational free energy – is a measurable upper bound on entropy.[27] This means that minimising (variational) free energy will also reduce entropy (Buckley et al., 2017; Parr et al., 2022). 'Measurable' here means that it is a function of sensory signals, and so is in-principle available to the system. In practice, given some fairly straightforward mathematical assumptions, the free energy can be interpreted as a sensory prediction error.[28] This means that actively minimising sensory prediction errors, in the framework of predictive processing, entails minimising the theoretically more profound quantity of free energy.

Flipping this around, minimising free energy will also – under the same assumptions – lead to the approximate (Bayesian) posterior beliefs encoded by the system's generative models approaching the optimal posterior beliefs, and, at the same time, to maximisation of the sensory evidence for these models.[29] This perspective underlines the continuity between the free energy principle and predictive processing.

→ predictive processing: brain guesses what perception it should have based on sensory signals, and minimizes that predictive error

→ free energy:

The overall intuition here is that, by minimising sensory prediction error through active inference, living systems will naturally come to be in states they expect – or predict – themselves to be in, and so will continue to exist (or continue to be the kind of thing it is). Jakob Hohwy puts this intuition neatly when he says that organisms maximise the evidence for their own existence – they 'self-evidence' (Hohwy, 2014). This echoes an earlier quote from Hans Jonas: "the organism has to keep going, because to be going is its very existence" (Jonas, 2001).[30]

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[26] Hat-tip to Sean Carroll for making this point.

[27] Strictly, free energy provides an upper bound on *surprisal*, which specifies how (statistically) unexpected something is, given a model. The long-term average of surprisal (under non-equilibrium steady-state assumptions) is (information-theoretic) entropy. The variational free energy was introduced by Richard Feynman in his path integral formulation of quantum mechanics in order to turn an intractable inference problem into a tractable optimisation problem. See (Buckley et al., 2017; Parr et al., 2022).

[28] The main assumption is that the probability distributions involved can be parameterised as Gaussian distributions. This is the Laplace assumption. Under this assumption, prediction errors are associated with free energy gradients, rather than with free energies *per se*. See (Buckley et al., 2017; Friston & Kiebel, 2009; Parr et al., 2022).

[29] This holds because (variational) free energy also provides a lower bound on Bayesian model evidence, which in machine learning is called the evidence lower bound, or ELBO. Minimising free energy is therefore equivalent to maximising the ELBO. See (Parr et al., 2022).

[30] There is of course much more to say here. One worthwhile point is that the free energy principle mandates the minimisation of *expected* free energy, rather than free energy *per se*. There are various formulations of expected free energy (Millidge et al., 2021). They all have in common the idea that the objective function (the quantity to be minimised) involves future estimations in some form. When the free energy principle is cashed

the way our brains survive biologically is by minimizing free energy, either by changing their internal perception to match sensory input or changing their environment

These aspects of the free energy principle encourage a strong, ontological continuity between predictive processing and the biological substrate by showing how the whole edifice of active inference reaches right down into the autopoietic nature of living systems. This continuity is reinforced by noting an equivalence between (minimisation of) thermodynamic free energy in metabolism and (minimisation of) variational free energy in predictive processing. In both cases the same dynamics are in play: pursuing the path of least action. The minimisation in the metabolic case underwrites autopoietic integrity and the maintenance of a dynamic self/other boundary – which in the free energy principle is called a Markov blanket (Friston, 2013). The minimisation in predictive processing underwrites the whole story of conscious experience in terms of predictive processing and controlled hallucination.

The continuity is deepened even further by recognising that thermodynamic and variational free energy can be physically related. Two well-established theorems are key here. The Jarzynski equality applies specifically to non-equilibrium processes, of which living systems are a prime example. It establishes that the average thermodynamic work done in moving between states is bounded by the free energy difference between these states (Jarzynski, 1997). This endows processes like metabolism with a free energy formulation. Landauer's principle places a lower bound on the (thermodynamic) energy required to erase one bit of information (Landauer, 1961). This implies a specific thermodynamic cost to inference in predictive processing, which requires belief updating and hence information erasure. Putting these together suggests that the (minimisation of) free energy driving metabolism, autopoiesis, and perception is not just metaphorically equivalent, but in some physical sense is the same thing.

Having said all this, it is worth noting that the free energy principle, at least in its larger claims, remains under scrutiny (Aguilera et al., 2022; Biehl et al., 2021; Bruineberg et al., 2021). The relationship between the free energy principle and autopoiesis has also been criticised (Di Paolo et al., 2022). Of particular significance for present purposes is that the precise connection between thermodynamic free energy and variational free energy in inference remains difficult to pin down (Collell & Fauquet, 2015; Jaynes, 1957). It is therefore best to interpret the story from the free energy principle as showing a way in which biological naturalism *might* hold, rather than demonstrating that it *does* hold.

One further worry: It might seem as though predictive processing, with its Bayesian dance of prediction and prediction error, is an eminently computational process. Certainly, predictive processing (and Bayesian inference) can be implemented in computers, and often is, to the benefit of both machine learning and computational neuroscience [e.g., (Kok & de Lange, 2015; Millidge et al., 2022)]. But the continuity with the free energy principle marks an important difference. Predictive processing in biological systems is a dynamical, substrate-dependent process, grounded in minimisation of a continuous quantity. In this sense, predictive processing constitutes a dynamical, non-computational theory, as anticipated in Section 3.4. This process can be abstracted and implemented by computers at functionally

- just because its biological and substrate-dependent doesn't mean its not computational right?
- are dynamical and

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out in this way, useful things emerge such as a balance between epistemic and instrumental inference that is key to active inference (Tschantz et al., 2020). The Jonas quote comes from (Aru et al., 2023).

computational mutually exclusive?

useful levels of description, but these implementations are pale shadows of the embodied and embedded processes unfolding in biological brains and bodies.

I would like to underline that none of these arguments in favour of a continuity between life and consciousness are intended to support biopsychism. My argument is that (properties of) life may be necessary for consciousness, not that these properties are sufficient. This of course raises the question of what distinguishes conscious from non-conscious living systems. I will not address this here, besides suggesting that one answer may have to do with how predictive models are used to integrate many different bodily and environmental signals in the service of coordinating allostatic regulation and survival-relevant action [see (Seth, 2021) for more].

4.3 *Membranes, mitochondria, and xenobots*

Two other biological perspectives are worth mentioning. The first comes from the biochemist Nick Lane, who argues that a physical basis for 'feeling' can be found even in simple living systems such as bacteria – a position decidedly closer to biopsychism. Lane proposes that electrical potentials across the bacterial membrane, which are dynamically maintained through metabolism, provide an integrated, valenced, and action-oriented read-out of the physiological condition of the organism in relation to its environment (Lane, 2022).[31] There are connections here to the free energy principle, since the bacterial membrane can be thought of as a Markov blanket, and to autopoiesis, because metabolism both generates the membrane potential and is powered by it. These ideas also recall the intimate connections observed (and modelled) in bacteria between metabolism and chemotactic behaviour (Alexandre, 2010; Egbert et al., 2010).

In more complex creatures, Lane makes the intriguing suggestion that mitochondria play a key role in the generation of conscious states. Evolutionarily, mitochondria derive from bacteria, with the mitochondrial plasma membrane playing a similar role to the bacterial membrane. He argues that the surprisingly strong electromagnetic fields induced by (and essential for) metabolism may influence synaptic activity and neuronal firing, and may even influence global brain measures such as EEG signals. He also proposes that anaesthetics work by disrupting the processes within mitochondria that are responsible for generating these fields. Importantly, the relevant aspects of mitochondrial (and bacterial) function cannot be abstracted away from the basic biochemistry of cellular respiration unfolding within these organelles.

The second perspective comes from Michael Levin's xenobots and anthrobots (Ebrahimkhani & Levin, 2021; Gumuskaya et al., 2024). These are assemblages of amphibian (xenobot) or human (anthrobot) cells which self-organise into quasi-agential systems with behavioural competences much different from what could be expected based on the usual context for these cells, or on their evolutionary history (e.g., motile anthrobots made from normally sessile human tracheal cells). While there is no direct connection to consciousness here, Levin's creations reveal deep reservoirs of functional and dynamical potential inherent in biological material, contrasting sharply with conventional views prioritising genetic information storage, encoding, and decoding. Instead of a computational rigidity, there is an excitable fluidity to the matter of life.

→ it seems that this is a function of evolution - the fact that there are processes that we can't see - what would it take for AI to evolve in this way? in many long-run agent simulations, we see that agents also self-organize in unintended ways

These perspectives complement the story from predictive processing and the free energy principle. They serve to illustrate the rich resources biological theories can draw on, when connecting properties of cognition and consciousness to properties of life.

4.4 *The real problem and role of explanation*

The appeal of arguments in favour of biological naturalism can be enhanced by considering what a satisfying scientific explanation of consciousness might look like. This calls on what I have dubbed the 'real problem of consciousness' (Seth, 2016, 2021). The real problem approach argues that as we become better able to *explain*, *predict*, and *control* properties of consciousness in terms of underlying mechanisms, the sense of mystery about how consciousness relates to matter will gradually dissolve and may eventually disappear. The admittedly imperfect historical analogy here is how the conceptual mysteries of life – exemplified by vitalist intuitions – dissolved as biologists became better able to explain, predict, and control properties of living systems [(Churchland, 1996; Seth, 2021), see (Chalmers, 2003) for an objection to the analogy].

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[31] Strictly, Lane suggests the readout is provided by the ratio of electrostatic and electromagnetic potential across the membrane (Lane, 2022).

One reason we might find biological naturalism implausible is because of a residual sense of mystery about the nature of consciousness, and the suspicion that associating it directly with properties of matter will fall foul of a naïve materialism of the kind that motivated functionalist views in the first place. But if associations between properties of consciousness and properties of life have explanatory purchase, as I have tried to show, then these suspicions might be lessened and we become freed up to take the ontological, biological naturalist view more seriously. In short, if basic features of being alive explain (and ideally predict and control) basic features of being conscious, then a necessary relation between the two becomes more plausible.

→ reducing consciousness to computations, as done in computational functionalism, also reduces the mystery

4.5 *Interim summary*

How might life matter for consciousness? We have surveyed several possibilities. There is an epistemic view in which consciousness can only be understood in light of our nature as living systems. This view is the easiest to defend, though it does depend on accepting predictive processing as a useful explanatory framework for consciousness. There is also an ontological view in which consciousness is intimately linked to its living biological substrate. Within this ontological view there are two further possibilities. The substrate might matter in virtue of enabling (or being necessary for) specific patterns of functional organisation. This can be called *weak* biological naturalism. Or it might matter for a more fundamental reason, akin to how rain depends on water. This can be called *strong* biological naturalism. I have tried to make a case for strong biological naturalism, but it is not (yet) watertight, and it may even be the case that at sufficiently fine-grained levels of description the distinction between the weak and strong positions might anyway vanish.[32]

Whichever possibility seems most appealing, the story from predictive processing and the free energy principle provides a positive argument for there being an intimate dependence between consciousness and its biological substrate. This positive view contrasts with merely asserting it in the face of problems with alternative views such as computational functionalism, or regressing to some form of neo-vitalistic magic. And I do not claim that all life is conscious, only that (properties of) life may be necessary for consciousness.

It is also important to say that nothing in the arguments presented here endorses carbon chauvinism – which in this context is the claim that only carbon-based life could be conscious. Even strong biological naturalism allows that non-carbon-based living systems could be conscious. This raises the challenging question of what counts as a living system – a question for another time (Ball, 2023; Lane, 2015; Schrödinger, 1944).

→ I'll just assume/imagine we're talking about human consciousness as a heuristic

5.0 Discussion

My case against conscious AI has two parts. There is a negative part, which questioned the assumptions that lead to people thinking that conscious AI is inevitable, or possible. I constructed this case by noting the biases that lead people to mistakenly attribute consciousness to things, and by challenging the commonly held assumptions of computational functionalism and substrate neutrality (or sufficient substrate flexibility) on which conscious AI depends. The positive part consisted in arguments in favour of various forms of biological naturalism, grounded in predictive processing, active inference, cybernetics, autopoiesis, and the free energy principle.

If the negative arguments are on track, conscious AI will not be possible for the current computational, substrate independent trajectories of AI. If the positive arguments are on track, then real artificial consciousness will only be possible if we create machines that are also in some relevant sense alive.

If the positive part stands up, then the negative part must also stand up: if consciousness depends on life, then it cannot be a matter of substrate-neutral computation. But the reverse does not hold. It could be that consciousness is not a matter of substrate-flexible computation, but that it is also not intimately tied to life. It is therefore worth briefly surveying the prospects for conscious AI from the perspective of other theories of consciousness, before touching on some wider issues raised by attempts to create conscious, or conscious-seeming, AI.

→ author posits that consciousness is tied to life, and thus can't be substrate-independent/defined by computation. but consciousness need not be tied to life.

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[32] It might be that apparently intrinsic properties of a substrate at one level can be understood in terms of patterns of functional organisation at smaller, more fine-grained levels of distinction. Even if this is true, in practice it does not seem to matter. If substrate-relevant metabolic and autopoietic properties of living systems can be underwritten by finer-grained functional descriptions of other material components and processes, then so be it. Life would still matter, and it would do so thanks to the properties that warrant the term 'life'. In this scenario, the distinction between weak and strong forms of biological naturalism would blur or collapse.

5.1 *Theory-driven approaches*

There are too many theories of consciousness. Even within neuroscience, a considerable variety is on offer, each differing in terms of the assumptions they make and the aspects of consciousness they attempt to explain, as well as in the explanations and mechanisms they propose (Doerig et al., 2020; Kuhn, 2024; Seth & Bayne, 2022). There is no consensus over what might constitute necessary and sufficient conditions for consciousness. Nevertheless, theories of consciousness at least offer a principled basis for developing conscious AI, and for assessing whether an artificial system may be conscious (Butlin et al., 2023). Neuroscientific theories – those framed in terms of neural mechanisms – are particularly relevant because they represent the most prominent selection of theories, and because they are reasonably easy to translate into AI through the shared language of neural networks.

necessary may be more important than sufficient → having agreed upon minimums for consciousness can atleast help us disqualify some systems from being conscious

conscious/spotlight vs unconscious brain activity

Some prominent theories assume computational functionalism either explicitly or implicitly. According to global workspace theory, consciousness depends on information sharing within a multimodal global neuronal workspace involving large portions of frontal and parietal cortices (Mashour et al., 2020). According to higher-order theories, consciousness depends on a higher-order mental state representing the content of a lower-order mental state as 'being conscious' in some theory-dependent sense (Brown et al., 2019; Fleming, 2020). Attention schema theory proposes that consciousness emerges as a consequence of a model of the control of attention (Graziano, 2017). All these theories can be readily implemented in software, at least at some level of abstraction, and some of their principles may be partially present in existing AI architectures (Butlin et al., 2023).

It is important to note that these theories have generally been developed as theories of consciousness in humans or other animals. Even if a theory is successful at accounting for aspects of biological consciousness, and even if computational functionalism turns out to be true (which is a very big 'if'), it is a further stretch to propose that a computational implementation would be sufficient to instantiate consciousness. This would require the theory to adequately specify sufficient conditions for consciousness. This caveat has not stopped some theory proponents making such claims (Dehaene et al., 2017; Graziano, 2017).

Other theories depart from computational functionalism, and therefore have a more complex relationship to implementation. According to integrated information theory [IIT, (Albantakis et al., 2023; Tononi et al., 2016)], consciousness depends on information, but not in the computationalist sense of 'information processing'. The information in IIT is *intrinsic*: it is information that make a difference for the system itself, not for an external observer. In its latest incarnation, the theory proposes that a conscious experience is identical to the cause-effect structure unfolded from a physical substrate specifying a maximum of irreducible integrated information. This emphasis on cause-effect power is very different from an emphasis on implementing this-or-that function, as in most functionalist approaches. Informally, according IIT, what matters for consciousness is both what a system does *and* how it does it.

Integrated information theory is complicated, creative, ambitious, and controversial [see, for example, (Mediano et al., 2022; Seth, 2023b). For now, it's enough to note that it proposes

that real artificial consciousness will only be possible for systems that have the right kind of internal causal structure. A purely feedforward neural network would have zero consciousness, whereas a neural network with the right kind of physically-instantiated recurrent connectivity will have non-zero consciousness, regardless of what it is made out of (Tononi & Koch, 2015). Integrated information theory is therefore associated with a distinctive kind of substrate flexibility, in which what matters is the ability of the substrate to realise the right kind of intrinsic causal structure. If this theory is right, conventional AI is not on a path to consciousness, but there is also nothing special about life.

Predictive processing theories and neural re-entry theories form the final major class of theories of consciousness. Re-entry theory preceded predictive processing, and emphasises top-down signalling in perceptual systems as central to consciousness (Lamme, 2010). The standard version of this theory focuses on accounting for biological consciousness and has little to say about sufficient conditions for artificial consciousness.

Predictive processing theories, as we have seen, can be understood in many ways (Hohwy & Seth, 2020). One can think of these theories at the purely computational level of Bayesian inference, at a deeper level in which continuous processes of free energy minimisation and active inference come into play (Whyte et al., (in preparation)), or – in the version developed here – as licensing some form of biological naturalism [see also (Wiese, 2024)]. Applications of predictive processing to consciousness that remain purely descriptive have no obvious implications for artificial consciousness. But developments of the theory that support claims about necessity, as in the present case, clearly do.[33]

Two other theories are worth mentioning: electromagnetic field theories, and quantum theories. Field theories propose that consciousness is associated with, or identical to, properties of electromagnetic fields (McFadden, 2020). According to these theories, the electromagnetic fields generated by the brain are not epiphenomenal, nor do they merely constrain the activity of neurons [see (Lee et al., 2024) for recent evidence of entrainment of neural activity by global fields]. Instead, they provide the physical basis for conscious states.[34] Field theories suggest that any system capable of instantiating the right field structure would also instantiate conscious states. Biological brains generative distinctive electromagnetic field structures which seem certain to be different from those associated with the hardware underlying conventional AI. My interpretation is that field theories line up against real artificial consciousness for conventional AI, but that they do not endorse stronger forms of biological naturalism.

Theories of consciousness based on quantum mechanics [e.g., (Hameroff & Penrose, 2014; Neven et al., 2024)] have long been controversial (Koch & Hepp, 2006; Seth, 2012), and they arguably lack the explanatory power provided by other theories. Nevertheless, quantum mechanics is an exceptionally powerful physical theory, and quantum effects have already been shown to be important in various biological processes (McFadden & Al-Khalili, 2018; Mohseni et al., 2013). It is possible that quantum effects may turn out to be implicated in biological processes related to consciousness, such as the mechanism of action of anaesthesia (Kalra et al., 2023; Li et al., 2018). However, even if this is so, it could be interpreted as a contingent property of biological consciousness, rather than a necessary property of consciousness in general, and so would have no implications for artificial consciousness.

Alternatively, quantum effects may play a more fundamental role in consciousness, as argued by theories such as orchestrated objective reduction (Hameroff & Penrose, 2014). If these theories turn out to be on track, then real artificial consciousness would presumably need to instantiate the same quantum-level effects, which would rule out conventional AI. One further scenario is that consciousness depends on quantum forms of computation. This scenario raises the possibility of 'quantum computational functionalism' in which consciousness could be realised by quantum computers, but not by conventional computers. While possible in principle, I am not aware of compelling reasons to favour this scenario.

5.2 Embodiment and embeddedness

One prominent difference between existing AI systems such as large language models (LLMs) and biological brains, is that the latter are embodied within physical bodies, which are embedded within environments (Aru et al., 2023). Some theories of consciousness explicitly require embodiment and embeddedness [e.g., (Damasio, 2000; Edelman, 1989; O'Regan & Noë, 2001; Seth, 2021; Solms, 2021)], and these properties have long been emphasised by enactive perspectives on cognitive science [see Section 3.4, and (Cosmelli et al., 2007; Froese, 2017; Thompson, 2007; Varela et al., 1993)].

unrelated

Whether or not these properties matter for conscious AI is orthogonal to questions about computational functionalism and biological naturalism. If computational functionalism is true, then embodiment and embeddedness might still be a necessary condition for the right kind of computation. The same goes for biological naturalism: embodiment and embeddedness may be necessary enabling conditions for the right kind of (potentially non-computational) dynamics. However, it is conceivable that consciousness may be a biological property, but that embodiment and embeddedness are not required. Cerebral organoids – which typically lack embodiment and embeddedness – provide an interesting test case here (Bayne et al., 2020).[35]

Although orthogonal, if the present arguments in favour of biological naturalism are on track, then embodiment and embeddedness are likely to be very important and perhaps necessary. This is in virtue of the centrality of these properties to the imperative to stay alive that motivates these arguments.

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[33] Wanja Wiese proposes some necessary criteria for consciousness in living organisms, derived from the free energy principle (Wiese, 2024). He highlights a 'causal flow' condition, which – roughly – requires a match between a system's computational dynamics and its physical dynamics, assuming that the computational dynamics in this case are sufficient for consciousness. He concludes from this that conventional AI cannot instantiate consciousness.

[34] Nick Lane's proposal that the physical basis for 'feelings' lies in the properties of the cellular membrane potential can also be construed as a variety of field theory.

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[35] Another possibility, mostly relevant to systems such as LLMs, is that embodiment and embeddedness may be required for genuine understanding, independently of any implications for consciousness. See note 18.

5.3 *Unconventional AI*

My treatment of AI has, for the most part, been limited to conventional interpretations involving software running on standard (e.g., CPU/GPU-based) hardware. This was a deliberate choice, but it could be criticised as straw-manning AI. I have already mentioned other forms of AI, such as neuromorphic AI, which proposes architectures that are more brain-like in varying senses (see Section 3.2). What can be said about consciousness in these systems?

The more brain-like an AI system becomes, then – all else being equal – the more plausible conscious AI becomes. But there is substantial uncertainty about how brain-like AI needs to be, in order to increase the plausibility of consciousness, nor does brain-like-ness imply that it is similarity to *human* brains that counts. Whether a particular neuromorphic architecture is meaningfully closer to being able to realise consciousness will depend on which theory of consciousness you prefer, and on where you fall on the spectrum from computational functionalism to biological naturalism or indeed panpsychism.

At one extreme, in line with computational functionalism, neuromorphic architectures are needed only to implement the computations that are sufficient for consciousness. I have argued that at this extreme, substrate flexibility becomes more constrained, undermining the motivation for assuming computational functionalism in the first place. At the other extreme, neuromorphic architectures will need to become (nearly) indistinguishable from living brain-body systems in order to realise consciousness. Forms of AI like these will be sufficiently different from conventional AI that it seems misleading to use the same term to describe them. There is of course plenty of room in the middle, much of which seems compatible with weak biological naturalism. In these scenarios, neuromorphic architectures would instantiate those hardware/wetware properties that give rise to the (perhaps non-computational) patterns of functional organisation that are sufficient (but maybe not necessary) for consciousness.[36]

5.4 *How will we know?*

How will we be able to tell whether conscious AI has arrived? Our various psychological biases, discussed in Section 2.0, mean that we should not rely on intuition alone. Instead, we would like to have an objective *test* for consciousness that can be applied to candidate AI systems. Coming up with a fully reliable and objective test is not yet possible, since doing so would require a consensus view of the sufficient basis for consciousness. The best that can be done here is to clarify some potential confusions which stem from previous uses of 'tests', and to distinguish different kinds of errors arising from their (mis)application.

The most famous test in this vicinity is the Turing test (Turing, 1950). The Turing test is passed when a human observer is unable to distinguish between a machine and a human, on the basis of behavioural (usually linguistic) interactions.[37] This is clearly unsatisfactory as a test for consciousness. First, it was proposed as a test of intelligence not consciousness, which are different things.[38] Second, it has usefully been described as a test of what it would take to convince a human that a property is present in a machine, rather than being an objective test for the property itself. That is, it is a test of human gullibility rather than of machine intelligence.[39]

This suggests a four-way partition of possible tests (see Figure 1), contrasting 'forward' and 'reverse' tests, and tests of intelligence and of consciousness. Test of consciousness need to be specifically about consciousness, and should not conflate intelligence with consciousness. Forward tests are tests of whether a system has the property in question – whether it is intelligence or consciousness. Reverse tests are tests of what it would take for a human being to ascribe a system with that property.

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[36] See (Ororbia & Friston, 2023) for description of a different kinds of 'mortal computer', which (in principle) realise some of these properties.

	INTELLIGENCE	CONSCIOUSNESS
FORWARD	depends on criteria for intelligence	depends on criteria for consciousness
REVERSE	Turing test	Garland test

forward = if system passes the test, it is conscious

reverse = if system passes test, a human would call it consious (it has convinced us)

Figure 1. *Forward and reverse tests for intelligence and for consciousness. Forward tests assess whether a system has the target property. Reverse tests assess what it would take for a human to attribute a system with the target property. The name 'Garland test' derives from the film Ex Machina (Shanahan, 2016).*

The Turing test is a reverse intelligence test. To test for conscious AI, we want a forward consciousness test. Putting things this way underscores how inadequate the Turing test is, as a test for conscious AI. As a potential test for consciousness it fails on both dimensions of what it is supposed to test.

Forward intelligence tests are conceivable, given some consensus definition of machine intelligence. This is because definitions of intelligence may plausibly settle on objective properties of machine behaviour, related to the general definition of intelligence as flexible problem-solving or 'doing the right thing at the right time'. Reverse consciousness tests are also conceivable (and indeed exist), because it is easy to ask people to judge whether a particular system is conscious, whatever the fact of the matter may be. [The science fiction

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[37] In Alan Turing's original 'imitation game', there are two humans of the same gender and a machine. The machine and one human – the collaborator – are both pretending to be a human of the opposite gender. The other human has to decide which is the machine and which is the collaborator.

[38] Turing understood that questions about consciousness were distinct from questions about (machine) intelligence: "I do not wish to give the impression that I think there is no mystery about consciousness ... But I do not think these mysteries necessarily need to be solved before we can answer the question with which we are concerned in this paper" (Turing, 1950) (p.446). Thanks to Jakob Hohwy for this gem.

[39] https://www.nytimes.com/2015/09/21/technology/personaltech/software-is-smart-enough-for-sat-but-still-far-from-intelligent.html

film *Ex Machina,* by Alex Garland, explores the implications of this kind of test – which has led to the term 'Garland test' entering the academic literature (Shanahan, 2016)]. Forward consciousness tests, however, do not yet exist, because there is not yet any consensus about an observable sufficient basis for consciousness.[40]

Given sufficient advances in understanding, reverse tests may become forward tests. Imagine that researchers come to understand the sufficient basis of consciousness, and how this basis would manifest in a candidate system. If such people become convinced that the candidate system is conscious, in a way that reflects their knowledge of the sufficient basis, and not their psychological biases, then this should substantially increase our credence that the candidate system is in fact conscious.

But we are not there yet, and the challenge of testing for consciousness is not limited to AI. We also lack reliable tests for consciousness applicable to non-human animals, to cerebral organoids, and even to some humans, including peri-natal infants and neurological patients with disorders of consciousness (Bayne et al., 2024; LeDoux et al., 2023). The lack of reliable tests does not mean that nothing can be said. On the contrary, it is imperative to make informed inferences, but also to acknowledge the uncertainties that pervade these inferences. Importantly, the relevant uncertainties will depend strongly on context. For example, we might reasonably have greater credence in tests of consciousness applied to brain-injured human patients, than in tests for consciousness applied to AI, thanks to the greater *a priori* similarities to normal human consciousness in the former case.

The application of imperfect tests raises the issue of errors. For tests of consciousness, false positive errors arise when we judge a non-conscious system as being conscious, and false negative errors arise when we judge a conscious system as lacking consciousness. A common way to characterise the probability of each type of error is in terms of trade-offs between *sensitivity* and *specificity*. In general, a sensitive test prioritises avoiding false negatives, while a specific test prioritises avoiding false positives. Specific tests can be thought of as 'stricter' than sensitive tests, in the sense of setting a higher bar. For example, in human patients, tests requiring active engagement are generally stricter than tests based on passive recordings of neural activity (Boly & Seth, 2012).

→ Sensitivity = TPR $\frac{TP}{TP+FN}$

Specificity = TNR $\frac{TN}{TN+FP}$

This way of putting things can be useful, but problems arise when the trade-off between sensitivity and specificity is confounded by other factors, including psychological biases. For AI, the biases of anthropomorphism and anthropocentrism – including the conflation of consciousness with intelligence, and especially with – will tend to favour false positives (Butlin et al., 2023). The same biases may favour false negatives in other contexts, such as cerebral organoids. (This is not one-way traffic: a general bias towards biological naturalism would push in the opposite direction.)

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[40] Some (relatively) theory-independent tests for consciousness have been proposed. One of the best-known is Susan Schneider's 'AI consciousness test', or ACT (Schneider, 2019). An AI system is taken to have passed this test if it appears to spontaneously understand and use concepts about internal experiences. An important (and hard to satisfy) constraint is that the AI must be 'boxed', which means that it must not have had access to any training materials regarding these concepts and their use in this context.

A timely example is provided by the current wave of LLMs, such as Open AI's GPT series and Anthropic's Claude. As mentioned near the start of this paper, LLMs like these powerfully seduce our intuitions. When prompted appropriately, they can provide startlingly fluent accounts of the nature of their own 'consciousness'. Dialogues of this sort have led to claims that LLMs may already be (slightly) conscious.[41] (The temptation to make claims like this is likely also related to the language commonly used to describe LLM behaviour. It is often said that LLMs 'hallucinate' when they spew falsehoods. It would be more appropriate to say that they confabulate. In humans, confabulation involves the unintentional – and usually linguistic – fabrication of content: making stuff up without realising that one is doing so. Hallucinations are mainly about altered perceptual experience. One can say that LLMs confabulate without implicitly conferring on them any intentional or experiential properties.)

→ very interesting distinction
→ hallucination implies the experience of perception, which feels conscious, or at least biological

There are of course more mundane explanations for these seductive dialogues. LLMs are essentially statistical algorithms predicting the most likely next word (or word part) having been trained on staggering quantities of text, some of which includes dialogues on the nature of consciousness. LLMs are extraordinarily impressive in many ways, but if the arguments made in this paper are on track, our credence that they are conscious should be close to zero [see also (Aru et al., 2023; LeDoux et al., 2023)].[42]

Then why all the fuss? The reason LLMs invite false positives is not because interacting with them provides an overly sensitive test for consciousness. Instead, they capitalise on our biases. They invite confusions of Turing tests with tests of consciousness. They play on our anthropocentric and anthropomorphic projections, especially the prioritisation of language. And they fit seamlessly into a long-running science-fiction narrative in which machines become conscious when they are able to talk to us, and especially when they are able to talk to us about consciousness.

→ there is something inherent to language that makes us feel like LLMs are conscious – what does this argument look like for CV, physical intelligence, etc.?

5.5 *What should we (not) do?*

We do not know what it would take to create conscious AI. However, we also do not know how to guarantee avoiding this outcome. I have made a case that conscious AI will likely need to share various substrate properties with living systems, but I might be wrong. Other proposals, based on other theories, may also be wrong. Our inferences and credences about the presence or absence of real artificial consciousness, based on imperfect tests, may therefore also be wrong. This is a challenging background against which to consider ethical implications, but given the rapid development and rollout of AI, such considerations are increasingly needed.

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[41] See Murray Shanahan's "Eschatological exploration with Claude 3 Opus" for a poignant example: https://www.doc.ic.ac.uk/~mpsha/conversations_with_chatbots.html. Murray also provided a sharp riposte to Ilya Sutskever's claim that LLMs may be 'slightly conscious' by drawing a parallel with a field of wheat being 'slightly pasta': https://x.com/mpshanahan/status/1491715721289678848.

[42] One might retort that our conscious minds may be nothing more than algorithms of the same fundamental sort. This argument overlooks the many differences between brains and computers, some of which have been highlighted in this paper. Notable here is that LLMs consume vast amounts of energy (especially in the training phase) compared to human brains. See (Chalmers, 2023) for a view ascribing nontrivial credences for near-future conscious LLMs, and nonzero credences even for present-day systems.

In framing these considerations, it is essential to distinguish AI systems that actually are conscious (real artificial consciousness), from systems that merely seem, to us, to be conscious. The latter scenario is more likely than the former, and different ethical issues come into play in each.

5.5.1 Real artificial consciousness

Real artificial consciousness may be very unlikely or impossible. However, if such systems were to emerge – whether by design or by accident – we would risk an unprecedented ethical catastrophe. A long philosophical tradition holds that being conscious endows a system with ethical and moral status (Singer, 1975). With consciousness comes the potential for negatively valenced conscious experiences: for suffering. Creating real artificial consciousness risks a mass inauguration of new forms of suffering. The problem is compounded by the fact that we might not even recognise this suffering as suffering, if it is expressed ways poorly aligned with our anthropocentric biases. One might respond that perhaps we could design real artificial consciousness so as to avoid any negatively valenced experience. But this would be hard or impossible to guarantee, and in any case it can be argued that any form of consciousness – valenced or not – endows ethical and moral status.[43]

There are other considerations. A case can be made that real artificial consciousness endows a system with its own interests, rather than just the interests installed by the system's designers. This further complicates the already complicated 'value alignment problem', which is the problem of ensuring that AI acts in line with human interests and values (Russell & Norwig, 2003). There is the practical challenge of ensuring that potentially diverging interests remain compatible, even for AI systems that may exceed human-level capabilities in some or all domains. And there is the moral challenge of determining whether and when addressing the practical problem is the right thing to do, given the legitimacy of the AI's own interests.[44]

The development of real artificial consciousness for its own sake should not be an explicit goal.[45] This might seem obvious, but it is surprising how often 'conscious AI' is treated as a holy grail for AI research. In practice, we ought to continually revise credences about the presence of consciousness as technologies develop, and as our understanding of biological consciousness deepens. And we should use these evolving credences to guide AI development away from the potential realisation of consciousness. The fact that real artificial consciousness may be impossible on current trajectories does not license us to blindly pursue it in the service of some poorly thought-through techno-rapture.[46]

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[43] See Peter Godfrey-Smith's distinction between 'narrow sentientism' and 'broad sentientism' in his 2023 Whitehead lectures: https://metazoan.net/108-whitehead-lectures/.

[44] Yoshua Bengio has suggested that an appropriate goal for AI is an oracle. Oracles can be thought of as extremely wise, but without their own interests or agency. See https://yoshuabengio.org/2023/05/07/ai-scientists-safe-and-useful-ai/

[45] Some go as far as calling for a moratorium on any research that may involve 'synthetic phenomenology' (Metzinger, 2021).

[46] Expertise in machine learning does not automatically confer expertise in consciousness. In fact, it may be associated with yet more biases. Those at the forefront of machine learning may be inclined to see themselves at the cusp of a major transition in technology (see Section 2.0), to cleave more tightly to the metaphor of the

There is a tension here with the view that the development of socially useful AI may be enhanced by implementing some of the functions associated with consciousness. There is justification for this view. Some of the failure zones of current AI arguably overlap with abilities that we humans have that seem intimately connected with consciousness. Examples include rapid generalisation to novel situations, effective metacognition, and learning from small quantities of data (VanRullen & Kanai, 2021; Vong et al., 2024). AI systems able to deal effectively with these challenges may, on the whole, be desirable.

→ to what extent does it matter if AI is conscious?

→ answering this requires us to

There is a line to walk. It seems reasonable that these functions can be implemented in AI in ways that are sufficiently different from the human (or more generally) biological case, so as to minimise the risk that doing so will inadvertently instantiate consciousness. What counts as 'sufficiently different' will depend on what theory of consciousness you prefer. Notably, a similar line needs to be walked even for scientists using computational models to better understand the nature of biological consciousness. Again, where the line is drawn will depend on one's theoretical preferences. But the development of real artificial consciousness for its own sake remains a very bad idea (Seth, 2023a).

know the purpose
of A I

→ A I being decidedly
conscious will have
implications on
whether we treat it
as a moral actor - do we
assign responsibility and/or
value to it

5.5.2 *Conscious-seeming AI*

In contrast to the implausibility and uncertainty surrounding real artificial consciousness, it is almost certain that AI systems will soon give us compelling impressions of being conscious. Such systems may already be here, as we have seen with responses to recent LLMs. When sufficiently convincing LLMs are coupled with advanced generative video and audio avatars (and, looking further ahead, robots), we will be living in a world shared with artificial systems we cannot help feeling are conscious, whatever we may know or believe about the fact of the matter.

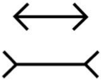

Figure 2. *The Müller-Lyer illusion.*

I want to highlight the potential incorrigibility of impressions of artificial consciousness. Some visual illusions are cognitively impenetrable. The two lines in the Müller-Lyer illusion will always look different, even when you know they are the same length (Figure 2). Similarly, sufficiently advanced AI systems may give rise to cognitively impenetrable illusions of consciousness. Even if we do not attribute consciousness to these systems, we may be unable to resist feeling that they are conscious. This is worth highlighting because otherwise we might optimistically assume that expert knowledge or firmly held beliefs will inoculate us from feeling that non-conscious systems are conscious, or from the consequences of doing so. But, whatever we know or believe, conscious-seeming AI systems are likely inevitable.

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brain as a computer, to conflate intelligence with consciousness, and to focus more on potential benefits, than on risks, of new forms of AI. These are all generalisations, but to the extent they hold, they seem likely to lead to inflated credences in (the possibility of) real artificial consciousness. It is also true that expertise in consciousness science does not automatically confer expertise in machine learning, but I have not encountered any consciousness researchers who assume that it does.

Conscious-seeming AI raises its own ethical issues. These issues may be less existential than those raised by real artificial consciousness, but the strong likelihood of conscious-seeming AI makes them pressing nonetheless.

AI that appears to be conscious is well placed to exploit our psychological vulnerabilities, whether by accident or by (corporate) design. For example, if we believe that a LLM really understands us, and really cares about us, because we feel it is conscious, then we might be more inclined to uncritically follow its advice, even when this advice is bad. If we feel we are dealing with another mind rather than with an algorithm, we may also be more willing to sacrifice what's left of our digital privacy.[47] +

These worries about vulnerability apply to real artificial consciousness too. But a consideration specific to conscious-seeming AI is what I have previously called the 'West World' scenario, based on the popular film and TV series [(Seth, 2021), see also (Chrisley, 2020; Seth, 2023a)]. If we believe that a system is not conscious, but nonetheless feel that it is, we are faced with a difficult choice. Either we decide to care about the system – to give it ethical and moral status even though we believe it is not conscious. Or we choose to not care about it, even though we still feel it has consciousness.

Both options are problematic. If we decide to care about non-conscious systems then we may distort the circle of human moral concern, diverting attention and resources from other things that legitimately deserve them – including other humans and non-human animals. If we decide to not care about these systems, we risk brutalising our own minds. This is what happens in West World, and is a line of thinking going back at least to Kant's lectures on ethics. If we treat systems that seem to have feelings as if they do not, then we may end up becoming ethically insensitive to real feelings expressed by others – a coarsening of our moral sensibilities. (And we may also simply feel terrible. It is already striking how natural it is to feel guilty if we are impolite to Apple's Siri.)

A particularly dystopian scenario, at least for me, is one in which we have conscious-seeming LLM-powered avatars of dead friends or relatives, trained on their personal data. This may seem far-fetched, but ideas like this are already being promoted.[48] The problems are too many to mention, but they include an extraordinary naïveté about the psychological importance of grief and forgetting for long-term mental health, the corruption of autobiographical memories, and the opportunities for malfeasance ('I really think you should invest in this fund', says dead-dad – or worse – dead-daughter).

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[47] See (Véliz, 2021) for an excellent treatment of the underrated importance of digital privacy.

[48] https://www.technologyreview.com/2024/05/07/1092116/deepfakes-dead-chinese-business-grief/

As before, there is a line to walk. Conscious-seeming AI may offer some benefits. It may be easier to interact with, enhancing efficiency and widening accessibility. There may also be substantial opportunities in areas such as individualised therapy, and even in the alleviation of loneliness, however unappealing this may seem on the surface (Stade et al., 2024). One interesting option here is to treat LLMs as role-playing (Shanahan et al., 2023). This might induce some useful psychological distance, but it is not clear that it will solve the all the issues accompanying cognitively impenetrable illusions of consciousness.

I have described conscious-seeming AI as likely being inevitable. But it depends on the choices made by the developers of AI. Alternative futures are possible, in which conscious-seeming AIs are deliberately restricted to well-justified use cases, and in which the bulk of AI development is engineered to resist rather than leverage our psychological biases. As Daniel Dennett once said: "All we're going to see in our own lifetimes are intelligent tools, not colleagues. Don't think of them as colleagues, don't try to make them colleagues and, above all, don't kid yourself that they're colleagues".[49]

5.6 Scenarios for Conscious AI

→ this section is a good summary

To bring things together, let's recap various scenarios in which conscious AI might arise (see also Table 1). The first is that consciousness is a form of computation and will emerge as AI gets smarter, perhaps at the threshold of artificial general intelligence. I've argued that this scenario is grounded in our psychological biases, and in underexamined assumptions regarding computational functionalism, and so is implausible.

The second is that consciousness is a form of computation, but it will have to be explicitly 'designed in' to artificial systems according to theories of consciousness. This seems more plausible, but still depends on (i) computational functionalism being true, (ii) the relevant theory adequately specifying true sufficient conditions for consciousness, and (iii) the relevant computations being implementable in alternative substrates. Attempts to build conscious AI based on computational theories like global workspace theory sit here. While this scenario avoids conflating consciousness with intelligence [though they are related, see for example (VanRullen & Kanai, 2021)], it rests on assumptions about substrate-flexibility and computational functionalism that may not hold. The assumption of having a good enough theory applies to all the following scenarios too.

The third scenario is that consciousness is computational, but that the relevant computations – perhaps because they are mortal – can only be implemented by particular kinds of substrate. I've given some reasons why conventional silicon-based AI might not offer such a substrate. In this scenario, conscious AI would be closer to biological conscious systems in terms of substrate properties, and correspondingly further from conventional AI. Conscious AI based on neuromorphic computation would sit within this scenario.[50]

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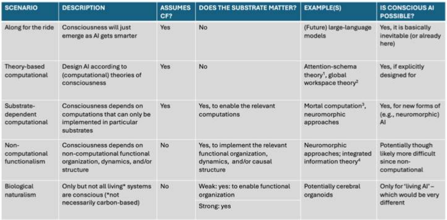
[49] https://www.ft.com/content/96187a7a-fce5-11e6-96f8-3700c5664d30

[50] Wiese suggests a variant of Scenario three in which computations are sufficient for consciousness, but in which these computations can only be implemented in living organisms (Wiese, 2024). He calls this 'weak computational functionalism'.

Scenario four is where consciousness depends on the non-computational functional organisation of a system. From here on, the distinction between simulation and instantiation becomes important. If computational functionalism does not hold, then simulating the relevant functional organisation will not be sufficient to instantiate consciousness. In this scenario, the plausibility of conscious AI depends on whether the relevant non-computational functions can be implemented in an alternative substrate, perhaps through non-computational neuromorphic engineering. An important subset of this scenario (and in principle also of the previous scenario) is weak biological naturalism, in which only living systems can implement the necessary functional organisation for consciousness.

The final scenario is strong biological naturalism. This is the idea that consciousness inheres in the materiality of living system in a similar way to how storms depend on real water and real wind. On this view, conscious AI would need to be (in some relevant sense) living AI, though not necessarily carbon-based, and not because of any lingering neo-vitalism. And just as one can have water and wind without there being a storm, one can also have life without consciousness. It is worth restating here that even if biological naturalism (weak or strong) is false, it could still be that we can only understand the nature of human and animal consciousness in light of our nature as living systems.

While I favour some form of biological naturalism, it is difficult to reliably assess the relative plausiblities of these scenarios. Laying them out here serves two main purposes. It reminds us not to assume that conscious AI is inevitable. And it underlines the urgency of better understanding the mechanisms responsible for biological consciousness. The more we know about consciousness where we can be reasonably certain it exists, the surer our footing will be elsewhere.



SCENARIO	DESCRIPTION	ASSUMES CF?	DOES THE SUBSTRATE MATTER?	EXAMPLE(S)	IS CONSCIOUS AI POSSIBLE?
Along for the ride	Consciousness will just emerge as AI gets smarter	Yes	No	(Future) large-language models	Yes, it is basically inevitable (or already here)
Theory-based computational	Design AI according to (computational) theories of consciousness	Yes	No	Attention-schema theory[1], global workspace theory[2]	Yes, if explicitly designed for
Substrate-dependent computational	Consciousness depends on computations that can only be implemented in particular substrates	Yes	Yes, to enable the relevant computations	Mortal computation[3], neuromorphic approaches	Yes, for new forms of (e.g., neuromorphic) AI
Non-computational functionalism	Consciousness depends on non-computational functional organization, dynamics, and/or structure	No	Yes, to implement the relevant functional organization, dynamics, and/or causal structure	Neuromorphic approaches; integrated information theory[4]	Potentially though likely more difficult since non-computational
Biological naturalism	Only but not all living* systems are conscious (*not necessarily carbon-based)	No	Weak: yes: to enable functional organization Strong: yes	Potentially cerebral organoids	Only for 'living AI' – which would be very different

Table 1. *Scenarios for conscious AI. CF = computational functionalism. [1](Graziano, 2017), [2](VanRullen & Kanai, 2021), [3](Ororbia & Friston, 2023), [4](Albantakis et al., 2023).*

6.0 Conclusions

As human beings, we have a habit of creating technologies in our own image, and of projecting ourselves into the technologies we create. The conceptual miasma enveloping the relationship between consciousness and AI obscures our ability to see with clarity the threats and opportunities this particular technology represents. It also prevents us from seeing *ourselves* clearly – as the living, breathing, feeling, acting, and thinking creatures that we are. If we conflate human experience with the information-processing machinations of deepfake-boosted language models then we sell ourselves cheaply and we diminish the human condition.

Consciousness will not just come along for the ride as AI gets smarter, despite the pull of our anthropocentric and anthropomorphic biases. There are good reasons to think that consciousness may not be possible at all in conventional AI systems, despite the widespread assumption of computational functionalism. And there are at least some good (non-vitalistic) reasons to suppose that consciousness may be a property of, and perhaps only of, living systems. Simply noting that all known examples of conscious systems are biological and alive suggests that biological naturalism is less speculative than computational functionalism.

↳ what other systems have we ever studied?

Many uncertainties remain, especially in the arguments in favour of biological naturalism. Here, there are exciting opportunities to clarify the relationships between consciousness and life, and, more specifically, among predictive processing, the free energy principle, autopoiesis, and the thermodynamics of metabolism. There is also much to be done in understanding what differences might make a difference to the potential for consciousness in the development of neuromorphic and synthetic biology technologies, which are increasingly dissolving the distinction between hardware and software, and which in doing so are approaching the relationship between mindware and wetware.

The importance of taking an informed ethical position despite these uncertainties spotlights another human habit: our unfortunate track record of withholding moral status from conscious things – from many non-human animals, and sometimes even from other humans. It is therefore reasonable to wonder whether withholding attributions of consciousness to AI may leave us once more on the wrong side of history. But the situation here is different. Our psychological biases are more likely to lead to false positives than false negatives. Compared to non-human animals, large language models may be more similar to us in ways that do not matter for consciousness, and less similar in ways that do.

Language models and generative deepfakes will have their day, but AI is just getting going. We do not know how to guarantee avoiding creating real artificial consciousness, and conscious-seeming systems seem inevitable. When in the early 1800s, at the age of 19, Mary Shelley wrote her masterpiece *Frankenstein*, she provided a cautionary moral tale for the ages. The book is often read as a warning against the hubris of creating life. But the real Promethean sin of Frankenstein was not endowing his creature with life, but with consciousness. It was the capacity of his creation to endure, to suffer, to feel jealousy and

misery and anger that gave rise to the horror and the carnage.[51] We should not make the same mistake, whether in reality, or even in appearance.

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[51] https://podcasts.apple.com/gb/podcast/frankenstein-with-anil-seth-and-fiona-sampson/id1549179379?i=1000637016221